Gradient Boosting with Decision Stumps

Swagatam Das

March 25, 2025

Toy Dataset

We have the following dataset:

Feature (X)	Target (Y)
1	0
2	0
3	1
4	1
5	1

Our goal: Predict the target Y based on feature X using Gradient Boosting with Decision Stumps.

Gradient Boosting - Initial Prediction

Initial Prediction:

- Start with the average of the target values for binary classification.
- $Y = [0, 0, 1, 1, 1]$
- The initial prediction for all data points is the mean of Y:

$$\hat{Y}_0 = \frac{0 + 0 + 1 + 1 + 1}{5} = 0.6$$

- So, the initial prediction for each sample is 0.6 (probability of class 1).

Step 2: Compute Residuals (Errors)

Now, we compute the **residuals**, which are the difference between the true target value and the predicted value.

$$\text{Residual}_i = Y_i - \hat{Y}_i$$

For each sample:

Feature (X)	**Target (Y)**	**Initial Prediction (0.6)**	**Residual (Error)**
1	0	0.6	-0.6
2	0	0.6	-0.6
3	1	0.6	0.4
4	1	0.6	0.4
5	1	0.6	0.4

Step 3: Fit the First Decision Stump

Next, we fit a **decision stump** to the residuals.

- A decision stump is a tree of depth 1, which splits the data based on one feature.
- Let's try a split at $X = 2.5$:
  - Group 1 ($X \leq 2.5$): $\{(X = 1, Y = 0), (X = 2, Y = 0)\}$
  - Group 2 ($X > 2.5$): $\{(X = 3, Y = 1), (X = 4, Y = 1), (X = 5, Y = 1)\}$
- The mean residuals for each group:
  - Group 1: $\frac{-0.6+-0.6}{2} = -0.6$
  - Group 2: $\frac{0.4+0.4+0.4}{3} = 0.4$

Step 4: Update Predictions

Now, we update the predictions using the decision stump's output, scaled by the learning rate (0.1).

Feature (X)	New Prediction	Residual
1	0.6 + 0.1 * (-0.6) = 0.54	-0.6
2	0.6 + 0.1 * (-0.6) = 0.54	-0.6
3	0.6 + 0.1 * 0.4 = 0.64	0.4
4	0.6 + 0.1 * 0.4 = 0.64	0.4
5	0.6 + 0.1 * 0.4 = 0.64	0.4

So, the new predictions are updated as shown.

Step 5: Compute New Residuals After First Stump

We now compute the new residuals based on the updated predictions:

Feature (X)	**New Prediction**	**Residual (New Error)**
1	0.54	0 - 0.54 = -0.54
2	0.54	0 - 0.54 = -0.54
3	0.64	1 - 0.64 = 0.36
4	0.64	1 - 0.64 = 0.36
5	0.64	1 - 0.64 = 0.36

Step 6: Fit the Second Decision Stump

For the second iteration, we fit another decision stump based on the new residuals.

- Try a split at $X = 2.5$ again:
  - Group 1 ($X \leq 2.5$): $\{(X = 1, Y = 0), (X = 2, Y = 0)\}$
  - Group 2 ($X > 2.5$): $\{(X = 3, Y = 1), (X = 4, Y = 1), (X = 5, Y = 1)\}$
- The mean residuals for each group:
  - Group 1: $\frac{-0.54 + -0.54}{2} = -0.54$
  - Group 2: $\frac{0.36 + 0.36 + 0.36}{3} = 0.36$

Step 7: Update Predictions (Second Iteration)

Now, we update the predictions using the second decision stump's output:

Feature (X)	New Prediction (After 2nd Stump)
1	$0.54 + 0.1 * (-0.54) = 0.49$
2	$0.54 + 0.1 * (-0.54) = 0.49$
3	$0.64 + 0.1 * 0.36 = 0.68$
4	$0.64 + 0.1 * 0.36 = 0.68$
5	$0.64 + 0.1 * 0.36 = 0.68$

Step 8: Final Predictions

After two iterations, the final predictions are:

Feature (X)	Target (Y)	Final Prediction
1	0	0
2	0	0
3	1	1
4	1	1
5	1	1

Final predictions: - $X = 1, 2$ predict class 0 - $X = 3, 4, 5$ predict class 1

Conclusion

In this hand-simulated example, we walked through:

- Two iterations of gradient boosting using decision stumps.
- How we computed the residuals and fit decision stumps to correct errors.
- How the model improves after each iteration by refining predictions.

This process continues until convergence or a set number of iterations is reached.